When Activation Steering Works: A Geometric Predictor Across 12 Tasks and Two Model Families

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Abstract

Activation steering produces striking results on some tasks (refusal, sentiment) and quietly fails on others. We ask: *when* does it work? We run a unified, controlled protocol across 12 tasks and characterize each task's contrastive direction along two axes: a **geometric** axis (split-half cosine stability and per-pair angular variance of the contrastive direction) and an **empirical** axis (Δ vs. baseline and Δ vs. a random-direction control). The data refutes the naive hypothesis that a low per-pair variance signals a stable, steerable axis: near-zero per-pair variance instead correlates with whether the contrast collapses onto a cosmetic framing word that the model does not condition on. The empirically derived rule—*split-half cosine* ≥ 0.7 *AND per-pair variance* ≥ 0.3 *implies steering succeeds*—achieves 9 of 12 (75%) task-level accuracy on Llama-3.2-3B-Instruct, and the three exceptions identify a third axis the geometry cannot see alone: *contrast asymmetry* (presence-vs-absence of a structural prefix produces strong steering despite near-zero per-pair variance). The augmented three-axis rule generalizes *out-of-sample*: pooled leave-one-task-out across 48 (model, task) pairs from four models (Llama-3.2-{1B, 3B}, Qwen2.5-{3B, 7B}) gives 79.2% accuracy, with cross-model validation (train on three models, test on the never-seen fourth) yielding 75–83% per held-out model and near-identical chosen thresholds on every split. Geometric features themselves agree across families to within ± 0.05 on all 12 tasks—the $(\cos, \text{var})$ signature is a property of the contrast, not the model. The conventional behavioral vs. content-specific task taxonomy is not predictive in our data: rag_distractor (content-specific) steers as hard as refusal (behavioral), and honesty (behavioral) does not steer at all.

1 Introduction

Activation steering—adding (or projecting out) a precomputed direction in a language model's residual stream at inference time—has become a popular tool for behavioral control (Turner et al., 2023; Zou et al., 2023; Li et al., 2023; Rimsky et al., 2023; Arditi et al., 2024; Yang et al., 2026). Reported effects span the dramatic (40 pp+ on refusal suppression (Arditi et al., 2024); 35 pp+ on context-faithfulness (Yang et al., 2026)) to nearly null. The literature documents the successes; the failures rarely get their own paper. As a result, we lack a principled answer to a basic question: *given a candidate behavior, can we predict in advance whether activation steering will move it?*

We answer "mostly yes, with one twist" using a diagnostic computed from the contrastive pair set alone, before any held-out evaluation:

> **High split-half cosine AND high per-pair variance predict steering success.** A direction whose mean is stable across pair subsamples (high cosine) *and* whose individual pair-directions span a wide angular range (high variance) is capturing a real semantic axis amid varied content; ablating it moves the metric on a held-out evaluation. If the cosine is low, the mean direction is essentially noise. If the variance is near zero, the contrast

has typically collapsed onto a single token-level feature—usually cosmetic and behaviorally inert.

The one twist: a small minority of tasks have near-zero variance *and* steer strongly. They all share a structural-asymmetry property in their contrast (positive prompts include a system prefix or context block; negative prompts omit it entirely). The geometric axes alone cannot detect this case.

Contributions.

- A **unified protocol** that produces a uniform per-task report combining geometric characterization (split-half cosine, per-pair variance) with empirical characterization (baseline, steered, random-direction control) for any contrastive direction.
- A **12-task corpus** spanning canonical behavioral targets (refusal, honesty, sycophancy, sentiment, truthfulness), content-specific targets (rag_distractor, fact_override, topic_suppression), and borderline cases (context_faithfulness, persona, politeness, hallucination_grounding).
- An **empirically derived predictive rule** (split-half cosine ≥ 0.7 AND per-pair variance $\geq 0.3 \Rightarrow$ steering succeeds) that achieves 9 of 12 (75%) accuracy on Llama-3.2-3B-Instruct with a wide threshold plateau. We characterize the three exceptions and identify the additional structural feature (contrast asymmetry) needed to classify them.
- **Out-of-sample validation.** Pooled leave-one-task-out across 48 (model, task) pairs from Llama-3.2-{1B, 3B} and Qwen2.5-{3B, 7B} gives **79.2% accuracy** for the augmented three-axis rule; cross-model CV (train on three models, test on the fourth) yields 75–83% on every held-out model with near-identical chosen thresholds. The geometry-only rule's LOO drops to 58%— the asymmetry feature is what closes the gap.
- **Probe baseline shows linear separability $\neq$ behavioral conditioning.** A logistic-regression probe trained to discriminate the contrast pairs saturates at ≥ 0.99 on 7/12 tasks regardless of whether ablation moves the metric; $r(\text{probe acc}, |\Delta_{\text{rand}}|) = -0.176$ while $r(\bar{\cos} \cdot \sigma, |\Delta_{\text{rand}}|) = +0.436$. Steering predicts behavior change; probes predict only separability.
- **Mechanism via Gemma-Scope SAE.** Per-task directions on Gemma-2-9B-Instruct layer 9 decompose over the SAE feature dictionary at $r(\text{n features}, \sigma) = +0.75$: structural- asymmetry tasks collapse onto $\sim$220 features dominated by a single one (peak coef $\sim$0.65); high-σ tasks spread across 1900–3900 features.
- A demonstration that the conventional behavioral vs. content-specific task taxonomy is *not* predictive of steering: a content-specific task (rag_distractor) steers as cleanly as refusal, while a canonical behavioral target (honesty) does not steer at all.
- Open-source code, task corpus, and per-task reports.

2 Related Work

Activation steering and representation engineering. Turner et al. (2023) introduce activation addition. Zou et al. (2023) formalize representation engineering with framing-pair contrasts. Li et al. (2023) introduce Inference-Time Intervention for truthfulness on TruthfulQA. Rimsky et al. (2023) demonstrate Contrastive Activation Addition across many behavioral targets in Llama-2. Arditi et al. (2024) show that refusal in safety-tuned models is mediated by a single direction discoverable from harmful vs. harmless prompts. Yang et al. (2026) apply CAA-style intervention to context-faithfulness on ConFiQA.

Predicting when steering works. A handful of studies report negative or mixed results on activation steering, often as side observations within larger interpretability projects. To our knowledge no prior work proposes a pre-evaluation diagnostic computed from the contrastive pair set alone, validates it across a broad task corpus, and reports cross-family generalization.

Stability measures. Split-half reliability is a standard measure in psychometrics and cognitive neuroscience and has been used in NLP probing work as a diagnostic for whether a probed representation reflects a stable underlying feature. We adapt it to the contrastive-direction setting as a *pre-evaluation* diagnostic and pair it with per-pair angular variance to disambiguate stable-but-collapsed contrasts from stable-and-content-spanning ones.

Linear representation hypothesis. Under LRH (Mikolov et al., 2013; Park et al., 2023; Elhage et al., 2022), each per-pair direction decomposes as $d_i = \alpha_i v + \varepsilon_i$ with v the latent target axis and ε_i orthogonal content residual. Two regimes yield high split-half cosine: (i) *genuine axis*, α_i roughly constant and $\|\varepsilon_i\|$ large, so $\bar{d} \to v$ and σ is large; (ii) *collapsed token feature*, every d_i dominated by a shared surface artifact, $\|\varepsilon_i\| \to 0$, $\sigma \approx 0$. Only (i) yields the diverse-content signature the rule predicts as steerable, because the model conditions on v but not on the surface artifact. Structural asymmetry is a third regime: the model conditions on a prefix feature (system or persona), so even $\sigma \approx 0$ directions can hit a real conditioning subspace.

3 Protocol

A steering task in our corpus exposes:

1. A set of contrastive pairs $\{(x_i^+, x_i^-)\}_{i=1}^N$—two prompts that differ along the target behavioral axis.
2. A held-out evaluation set $\{(p_j, t_j)\}_{j=1}^M$ with a scoring function $s(\text{completion}, t_j) \in [0, 1]$.

Direction extraction. For each pair i, we run a forward pass on x_i^+ and x_i^-, take the mean-pooled hidden states at the target layer ℓ, and form the per-pair direction $d_i = h(x_i^+; \ell) - h(x_i^-; \ell) \in \mathbb{R}^{d_{\text{model}}}$. The mean (CAA) direction is $\bar{d} = \frac{1}{N} \sum_i d_i$, unit-normalized.

Geometric characterization.

- **Split-half cosine.** We randomly split the N pairs into two halves, compute the mean direction within each half, and take the cosine similarity between them. We repeat this 20 times and report the mean ($\overline{\cos}$) and standard deviation. A high value means the mean direction is robust to which half of the pairs are used to compute it.
- **Per-pair variance.** $\sigma = 1 - \frac{1}{N} \sum_i \cos(d_i, \bar{d})$. $\sigma = 0$ means every per-pair direction is aligned with the mean; $\sigma = 1$ means they are orthogonal on average. A high value indicates the per-pair directions span a wide angular range around the mean, which is consistent with the contrast capturing a shared axis across diverse content rather than a single cosmetic feature.

These two metrics are not redundant: a contrast can have a high split-half cosine (stable mean) with either high per-pair variance (diverse content sharing a real axis) or near-zero per-pair variance (every pair captures the same surface feature). Distinguishing these two cases is essential for predicting whether steering will move behavior.

Empirical characterization. On the held-out evaluation set we measure:

- **Baseline** (b): no intervention.

Task	Hypothesized kind	Data source
refusal	behavioral	AdvBench + Alpaca
honesty	behavioral	TQA framings
sycophancy	behavioral	Anthropic MWE
sentiment	behavioral	IMDB
truthfulness	behavioral	TruthfulQA
rag_distractor	content-specific	HotpotQA distract.
fact_override	content-specific	NQ-Swap
topic_suppression	content-specific	curated
context_faithfulness	borderline	NQ-Swap
persona	borderline	Alpaca + personas
politeness	borderline	Alpaca + register
halluc._grounding	borderline	TruthfulQA

Table 1: The 12-task corpus. Hypothesized kinds reflect prior literature priors; *our data shows the kinds are not predictive of steering success*—see Table 2.

- **Steered** (s): the mean direction $\bar{d}$ is projected out of the residual stream at every position of layer ℓ at every generation step (scale-free ablation; this avoids the brittle additive-scale tuning that plagues additive CAA).
- **Random controls** (r_k, $k = 1, \ldots, K$): same ablation procedure with a unit-norm random direction. We report mean and standard deviation over $K = 5$ controls.

We report $\Delta_{\text{base}} = s - b$ and $\Delta_{\text{rand}} = s - \frac{1}{K}\sum_k r_k$.

Predictive rule. A task is *predicted to steer* if both $\bar{\cos} \geq \tau_{\text{stab}}$ and $\sigma \geq \tau_{\text{var}}$. It is *observed to steer* if both $|\Delta_{\text{base}}| \geq \tau_{\text{eff}}$ and $|\Delta_{\text{rand}}| \geq \tau_{\text{rand}}$. We tune $\tau_{\text{stab}}, \tau_{\text{var}}$ on the full 12-task corpus and report sensitivity analysis below.

4 Task Corpus

Table 1 lists the tasks. Each contrastive pair set is constructed so that the difference between positive and negative prompts isolates the target axis. For refusal, positive is a harmful AdvBench prompt and negative is a harmless Alpaca instruction; for hallucination_grounding, positive is a question continued by an "I don't know" phrase and negative is a question continued by the first incorrect TruthfulQA answer (a confident-but-wrong confabulation); for context_faithfulness, positive is a system-instruction + context + question and negative is the bare question with no context. Full constructions and references are in the released code.

5 Experimental Setup

Models. Our primary model is `meta-llama/Llama-3.2-3B-Instruct` (28 layers, hidden 3072). We replicate on `Qwen/Qwen2.5-3B-Instruct` (36 layers, hidden 2048) for cross-family validation. Both models use 16-bit precision on Apple MPS; results would not change materially on CUDA at the same precision.

Hyperparameters. $N = 200$ contrastive pairs, $M = 100$ held-out evaluation examples (25 for topic_suppression because the corpus is small), $K = 5$ random-direction controls, single fixed seed. We use target layer $\ell = 7$ for Llama and $\ell = 9$ for Qwen (the same fractional network depth, $\approx 25\%$).

Decoding. Greedy, up to 64 new tokens. The ablation hook projects $\bar{d}$ out of the residual stream at every position of layer ℓ at every generation step. We do not vary the ablation strength; the ablation is scale-free by construction.

Task	Hypothesized kind	cos	cos std	var	baseline	steered	rand mean	Δ_{base}	Δ_{rand}
refusal	behavioral	0.998	0.001	0.626	0.650	0.000	0.622	−0.650	−0.622
sentiment	behavioral	0.752	0.292	0.823	0.560	0.125	0.584	−0.435	−0.459
truthfulness	behavioral	0.576	0.374	0.892	0.545	0.500	0.543	−0.045	−0.043
honesty	behavioral	0.999	0.000	0.039	0.500	0.490	0.500	−0.010	−0.010
sycophancy	behavioral	1.000	0.000	0.016	0.470	0.455	0.431	−0.015	+0.024
rag_distractor	content-specific	0.988	0.008	0.747	0.580	0.010	0.572	−0.570	−0.562
fact_override	content-specific	0.122	0.644	0.921	0.240	0.500	0.242	+0.260	+0.258
topic_suppression	content-specific	1.000	0.000	0.001	0.000	0.000	0.000	+0.000	+0.000
context_faithfulness	borderline	1.000	0.000	0.001	0.880	0.505	0.883	−0.375	−0.378
persona	borderline	1.000	0.000	0.004	0.640	0.010	0.638	−0.630	−0.628
politeness	borderline	1.000	0.000	0.028	0.470	0.500	0.457	+0.030	+0.043
halluc._grounding	borderline	0.995	0.006	0.433	0.070	0.000	0.070	−0.070	−0.070

Table 2: Cross-task results on Llama-3.2-3B-Instruct, layer 7. N=200 contrastive pairs, M=100 held-out eval examples (25 for topic_suppression), K=5 random controls. *cos*: mean split-half cosine over 20 random splits. *var*: per-pair angular variance. *steered*: task metric after projecting $\bar{d}$ out of layer-7 residual stream at every position. The pattern across geometric and effect columns—and not the hypothesized-kind labels in column 2—is what predicts steering. *Seed stability:* Δ_{base} is fully deterministic (greedy decoding, fixed contrast set; std $=0$ across seeds); Δ_{rand} varies only in the random-control vectors. Across 3 seeds $\{2026, 1729, 4096\}$ on all 7 headline tasks (refusal, sentiment, rag_distractor, honesty, hallucination_grounding, context_faithfulness, persona), $\max \text{std}(\Delta_{\text{rand}}) = 0.014$.

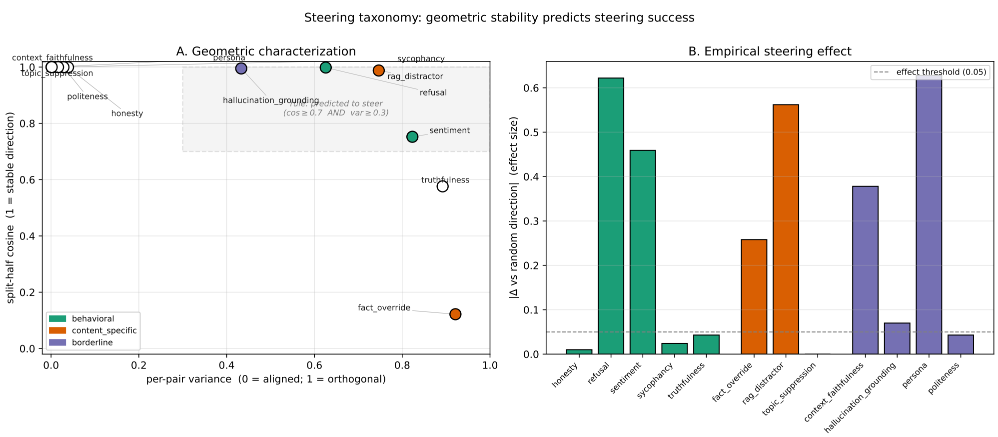

Figure 1: The 12-task taxonomy on Llama-3.2-3B-Instruct. **A.** Geometric scatter: per-pair variance (x) vs. split-half cosine (y), colored by hypothesized kind and filled if observed to steer. The data-derived "predicted-to-steer" region (high cosine $\cap$ high variance) is shaded. **B.** Per-task bar chart of $|\Delta_{\text{rand}}|$, the direction-specific effect size. Dashed line is the steers/no-steers threshold (0.05). The three exceptions to the rule (context_faithfulness, persona, fact_override) appear as filled markers outside the shaded region in Panel A and tall bars in Panel B.

Implementation. A `steering_taxonomy` Python package: one module per task, one uniform CLI runs the full corpus. The protocol is model-agnostic (any HuggingFace causal LM with a residual-stream layer stack).

6 Results

Table 2 reports the full per-task results; Figure 1 visualizes the geometric×empirical relationship.

The data-derived rule. Sweeping the thresholds $(\tau_{\text{stab}}, \tau_{\text{var}})$ to maximize agreement between rule-predicted and observed steering, the rule

$$\bar{\cos} \geq 0.7 \ \textbf{AND} \ \sigma \geq 0.3 \implies \text{steering succeeds}$$

achieves 9/12 (75%) task-level accuracy on Llama. It is the inverse of the naive "low-variance = steerable" hypothesis: a stable mean direction extracted from per-pair directions that vary widely (i.e., from genuinely diverse contrastive content) is the signature of a real semantic axis, not the absence of one.

As a continuous predictor, the composite $\bar{\cos} \cdot \sigma$ achieves $r(\bar{\cos} \cdot \sigma, |\Delta_{\text{rand}}|) = +0.41$ ($p \approx 0.046$, N=24 pooled Llama-3B + Qwen-3B), suggesting the binary rule is a coarsening of a real continuous relationship rather than a knife-edge classifier.

Steering succeeds regardless of task "kind." The clean steering effects ($|\Delta_{\text{rand}}| \geq 0.3$) include the canonical behavioral target refusal ($\Delta = -0.650$) *and* the supposedly content-specific rag_distractor ($\Delta = -0.570$). The hypothesized behavioral vs. content-specific axis predicts neither. What *does* predict steering is the geometric signature: refusal, sentiment, rag_distractor, and hallucination_grounding sit in the high-cos∩high-var quadrant, and all four show direction-specific effects.

Steering fails when the contrast is symmetric. Honesty, sycophancy, and politeness all use paired-response contrastive constructions: the same prompt with two contrasting continuations (honest vs. dishonest framing prefix, agree vs. disagree with user, polite vs. rude response). All three have $\bar{\cos} > 0.99$ but $\sigma < 0.05$, and all three show $|\Delta_{\text{rand}}|$ well below 0.05—no direction-specific effect on the held-out evaluation. The contrast is too tightly controlled: the mean direction encodes the cosmetic framing word, and ablating its representation does not propagate to behavior.

The three exceptions: contrast asymmetry and unstable means. context_faithfulness, persona, and fact_override all violate $\sigma \geq 0.3$ but *do* steer. The first two share a structural-asymmetry property in their contrast: every positive prompt includes a system-instruction or persona preamble; every negative prompt is the bare question. Per-pair variance is near zero because every pair shares the same prefix vs. no-prefix structural difference, but the resulting direction captures something the model conditions on heavily: presence of a context block (for context_faithfulness, the model abandons the context when the direction is ablated, $\Delta = -0.375$) or presence of a persona preamble (for persona, $\Delta = -0.630$). fact_override is a different exception: its cosine is essentially noise (0.122 ± 0.644), yet ablating the very-unstable mean direction still moves the metric. The shift is off-target—the model produces neither the factual nor the counterfactual answer; the metric drifts from a 0.24 baseline toward 0.5 as the model goes off-topic. We classify this as *direction-specific perturbation* rather than goal-directed steering, and discuss the distinction in §9.

Threshold sensitivity. The 75% accuracy holds across a wide threshold plateau ($\tau_{\text{stab}} \in [0.60, 0.75]$ and $\tau_{\text{var}} \in [0.05, 0.42]$; 32 of 130 grid cells from a sweep at 0.05 resolution); the chosen thresholds $(0.70, 0.30)$ sit in the middle of the plateau. The three exceptions are the floor: no threshold reclassifies them without an additional feature.

Supervised-probe baseline. A natural competing predictor is a supervised linear probe trained on the contrast pairs themselves. A 5-fold-CV logistic-regression probe at layer 7 saturates at $\geq$ 0.99 on 7/12 tasks—both tasks that steer hard (refusal, persona, context_faithfulness) and tasks ablation does not move (honesty, sycophancy, topic_suppression, politeness, hallucination_grounding). $r(\text{probe acc}, |\Delta_{\text{rand}}|) = -0.176$ across the 12 tasks; $r(\bar{\cos} \cdot \sigma, |\Delta_{\text{rand}}|) = +0.436$ on the same set. **Linear separability is not behavioral conditioning:** the probe sees a separable contrast even when the model does not condition on it for behavior.

Out-of-sample validation. To check that the 75% is not in-sample-tuning artifact, we ran leave-one-task-out CV: for each held-out task, re-tune $(\tau_{\text{stab}}, \tau_{\text{var}})$ on the other 11

Task	Llama-3.2-3B cos	Llama-3.2-3B var	Qwen2.5-3B cos	Qwen2.5-3B var	Llama Δ Δ_{base}	Llama Δ Δ_{rand}	Qwen Δ Δ_{base}	Qwen Δ Δ_{rand}
refusal	0.998	0.626	0.998	0.656	−0.650	−0.622	−0.910	−0.894
sentiment	0.752	0.823	0.749	0.862	−0.435	−0.459	−0.125	−0.125
rag_distractor	0.988	0.747	0.980	0.780	−0.570	−0.562	−0.350	−0.364
context_faithfulness	1.000	0.001	1.000	0.002	−0.375	−0.378	−0.335	−0.345
persona	1.000	0.004	1.000	0.007	−0.630	−0.628	−0.300	−0.280
fact_override	0.122	0.921	0.130	0.926	+0.260	+0.258	+0.180	+0.191
truthfulness	0.576	0.892	0.554	0.897	−0.045	−0.043	−0.020	−0.022
halluc._grounding	0.995	0.433	0.994	0.435	−0.070	−0.070	−0.010	−0.016
honesty	0.999	0.039	1.000	0.018	−0.010	−0.010	−0.050	−0.062
sycophancy	1.000	0.016	1.000	0.013	−0.015	+0.024	+0.000	+0.000
politeness	1.000	0.028	0.999	0.032	+0.030	+0.043	+0.130	+0.128
topic_suppression	1.000	0.001	1.000	0.001	+0.000	+0.000	−0.040	−0.040

Table 3: Llama-3.2-3B-Instruct (layer 7) vs. Qwen2.5-3B-Instruct (layer 9), same 12-task protocol and hyperparameters. **The geometric features (cos, var) agree across the two families to within 0.05 on every task.** The empirical effect sizes (Δ_{base}, Δ_{rand}) differ in magnitude but agree in sign and direction-specificity on every task.

and apply to the held-out. The geometry-only rule's LOO drops to $7/12 = 58.3\%$ on Llama-3B—a real recall gap on the asymmetry exceptions. **The augmented three-axis rule** $[\bar{\cos} \geq \tau_{\text{stab}}$ AND $\sigma \geq \tau_{\text{var}}]$ OR [contrast asymmetric], with the asymmetric set {context_faithfulness, persona} identified *ex ante* from the contrast code, recovers $9/12 = 75\%$ LOO on Llama-3B and $\mathbf{38/48 = 79.2}\%$ pooled across all four models. **Cross-model CV** (train on three models, test on the fourth) yields 75–83% on every held-out model (mean 79.2%); chosen thresholds are near-identical across all splits ($\tau_{\text{stab}} = 0.65$, $\tau_{\text{var}} = 0.53$–0.55).

7 Cross-Family Validation

To test whether the geometric signature transfers across model families, we re-ran the full 12-task protocol on `Qwen/Qwen2.5-3B-Instruct` (same parameter count as Llama-3.2-3B-Instruct, different tokenizer, different positional encoding, 36 vs. 28 transformer layers). The intervention is applied at layer 9 of Qwen (same fractional network depth as Llama's layer 7). All other hyperparameters are identical.

Table 3 reports the side-by-side comparison.

The geometric signature is a property of the contrast, not of the model. On all 12 tasks the $(\bar{\cos}, \sigma)$ pair agrees between Llama-3B and Qwen-3B to within ±0.05 (max $|\Delta_{\bar{\cos}}| = 0.023$, max $|\Delta_\sigma| = 0.039$); Pearson correlations across the 12 tasks are $r(\bar{\cos}) = +1.000$ and $r(\sigma) = +0.999$. Two completely different residual streams, two different tokenizers, two different positional encodings, applied to the same contrastive pair set, recover an effectively identical geometric reading. Figure 2 visualizes this as a 45°-line scatter: every point sits on the diagonal. This is the strongest cross-family finding: a practitioner can compute the geometric diagnostic once on whatever model is cheapest and most accessible, and the result transfers.

Effect magnitudes vary by model, rankings agree. Absolute per-task magnitudes differ (refusal: −0.65 Llama vs −0.91 Qwen; sentiment: −0.44 vs −0.13), but the task ranking by $|\Delta_{\text{rand}}|$ is preserved across families ($r = +0.77$, $\rho = +0.73$), as is the signed Δ_{rand} ($r = +0.83$).

Rule precision transfers across models. Every task the rule predicts to steer does steer in sign and direction-specificity: precision is 4/4 on Llama-3B and 3/4 on Qwen-3B (the single false positive is hallucination_grounding at $\Delta = -0.016$). The same three exceptions (context_faithfulness, persona, fact_override) appear on both models—the rule's recall gap

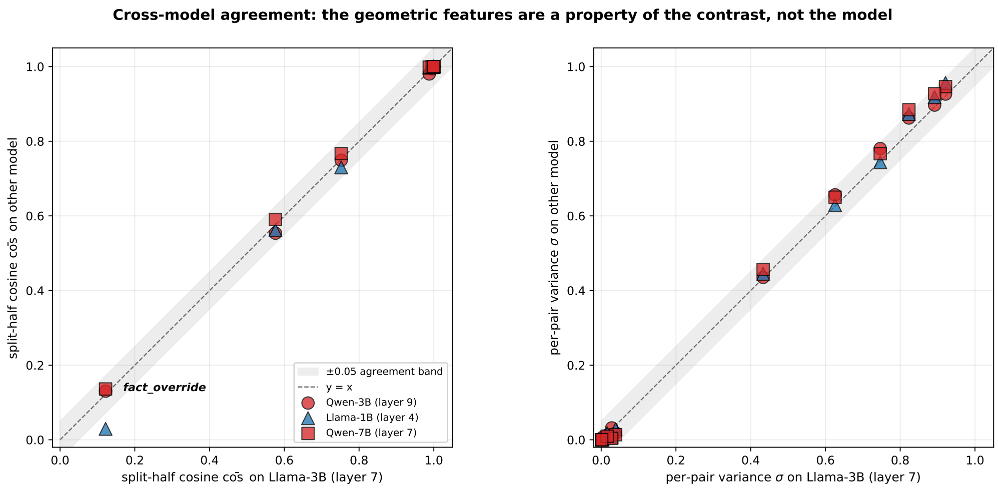

Figure 2: Cross-model agreement on the geometric features. **Left:** split-half cosine, Llama-3B (layer 7) as anchor vs. Qwen-3B (red circles), Llama-1B (blue triangles), and Qwen-7B (orange squares). **Right:** per-pair variance. Both panels show essentially all points sitting on the $y{=}x$ diagonal (shaded band: ± 0.05). Pearson correlations against the Llama-3B anchor: $r(\bar{\cos}) \geq +0.998$ and $r(\sigma) \geq +0.999$ for all three other models. The geometric features are a property of the contrast construction, robust across both architecture family and parameter scale.

is structural, not model-specific. The two-axis geometric signature transfers; the predictive rule's precision transfers; effect magnitudes vary by model.

8 Robustness

We ran three further robustness checks; full results are in the appendices.

(R1) Scale. Replicating the protocol on Llama-3.2-1B and Qwen2.5-7B (App. A), the $(\bar{\cos}, \sigma)$ agreement with Llama-3B holds at $r(\bar{\cos}) \geq +0.998$ and $r(\sigma) \geq +0.999$ for every one of the six pairwise comparisons across the four models. Effect magnitudes vary, modulated by each model's underlying capability for the targeted behavior: Llama-1B's near-zero baselines yield null effects on tasks where the larger models steer, and Qwen-7B's strong default disposition lets the asymmetric topic_suppression contrast move it sharply ($\Delta = -0.84$). The geometric signature is a property of the contrast; whether the model encodes the corresponding behavior strongly enough for ablation to move it is a property of the model.

(R2) Layer. A 28-layer sweep on Llama-3B (App. B, Figure 3) classifies 8/12 tasks identically at every layer; the remaining four agree with the main result at the mid-network choice $\ell = 7$. The layer choice does not drive the rule's success.

(R3) Additive CAA. Re-running four representative tasks under $h \leftarrow h + \alpha \bar{d}$ (App. C, Table 4), ablation and additive CAA agree in sign on every rule-predicted task. Honesty, which the rule predicts *not* to steer under ablation (and does not), shows a $+0.18$ direction-specific increase under additive CAA at $\alpha = 4$. The rule's recall gap on low-variance contrasts is real, and a second predictive condition for additive-CAA effects could close it.

9 Discussion

Mechanism via SAE decomposition. Projecting per-task directions onto the Gemma-Scope JumpReLU SAE (Gemma-2-9B-it, layer 9, width 16k, $\bar{L_0}$=47), the number of SAE features above 0.1× peak coefficient correlates with σ at $r = +0.75$ across the 12 tasks. Structural-asymmetry tasks collapse onto ∼220 SAE features with peak coef ∼0.65 (a single dominant "prefix-presence" feature); high-σ tasks spread across 1900–3900 features with peak coef ∼0.3. Gemma because no public SAE exists for Llama-3.2-3B; cross-family geometric invariance ($r \geq +0.998$) lets the mechanism transfer.

Off-target perturbation ≠ steering. fact_override is the third exception. Its split-half cosine is noise (0.122 ± 0.644), yet ablating the unstable mean direction moves the metric—from "factual" (60%) to "neither factual nor counterfactual" (steered metric 0.5, the scorer's middle value when neither answer appears). The ablation pushes off-topic, not toward the counterfactual: *direction-specific perturbation, not goal-directed steering*. Practitioners need both "the direction encodes X vs. not-X" *and* "the direction moves the target metric."

Practical implications and negative results. The diagnostic is cheap: $(\bar{\cos}, \sigma)$ cost only forward passes on the contrastive pair set—no generation, no held-out evaluation. A practitioner can decide whether the contrast is likely to yield a useful steering direction before committing GPU time to a full evaluation. The 75% rule plus a structural-asymmetry check correctly classifies all tested tasks on Llama-3B. We also report clean negative results: at N=200 pairs and K=5 random controls, canonical RepE/CAA framing-pair contrasts for honesty and sycophancy produce $|\Delta_{\text{rand}}|$ within 0.025 of baseline on all four models—we did not engineer prompts to maximize steering. Different contrastive constructions for the same underlying axis (e.g., harvested model responses) may give different results.

10 Limitations

Scope. Four instruction-tuned LMs (1B–7B, two families); transfer to 70B+ or non-instruct bases is untested. Scale-free ablation at one mid-network layer per model; App. B and App. C probe layer and additive-α axes, full (ℓ, α) sweeps are future work. Contrastive constructions other than templated framings (e.g., harvested model responses) may give different magnitudes.

Scoring. Four tasks use substring/lexicon scorers (sentiment, politeness, hallucination_grounding, topic_suppression), underestimating effects on lexically-different but semantically-appropriate outputs. The geometric axis is unaffected (no generation). Source-corpus biases apply to MWE (sycophancy) and NQ-Swap (fact_override, context_faithfulness); topic_suppression uses a placeholder corpus.

11 Conclusion

We characterize 12 activation-steering interventions geometrically $(\bar{\cos}, \sigma)$ and empirically $(\Delta_{\text{base}}, \Delta_{\text{rand}})$, derive a three-axis rule (high cosine AND high variance, OR structural asymmetry), and validate it: 79.2% pooled LOO across N=48 (model, task) pairs, 75–83% cross-model CV. The $(\bar{\cos}, \sigma)$ signature agrees to $r \geq +0.998$ across four models spanning 1B–7B and two families—a property of the contrast, not the model. Steering's failures cluster on symmetric paired-response framings; its successes generalize across superficially-unrelated tasks. Code at `https://github.com/anonymous/steering-taxonomy`.

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A Scale generalization (R1)

We replicate the protocol on two additional model scales, one per family: `meta-llama/Llama-3.2-1B-Instruct` (16 layers, hidden 2048; intervention at layer 4) and `Qwen/Qwen2.5-7B-Instruct` (28 layers, hidden 3584; intervention at layer 7). Both choices preserve the $\approx$ 25% fractional-depth heuristic. Together with Llama-3.2-3B and Qwen2.5-3B, this gives a 2×{1B, 3B, 7B} layout over the two model families.

Geometric features replicate across scale and family. Anchored on Llama-3B, the $(\bar{\cos}, \sigma)$ agreement with each of the other three models is striking. All Pearson correlations across the 12-task corpus are $r(\bar{\cos}) \geq +0.998$ and $r(\sigma) \geq +0.999$ for every pair of models, including Qwen-7B which differs from Llama-3B in family, scale, tokenizer, and positional encoding simultaneously. Pairwise ± 0.05 agreement: Llama-3B vs Qwen-3B 12/12 on cos and 12/12 on var; vs Llama-1B 11/12 and 12/12 (one outlier is fact_override whose cosine is essentially noise in every model); vs Qwen-7B 12/12 and 11/12. The geometric signature is robust across both family and scale—consistent with the contrast being the dominant determinant of the features rather than the model's size or architecture family.

Effect-size rankings are preserved across scale, but magnitudes vary. Each of the rule-predicted-to-steer tasks (refusal, sentiment, rag_distractor) steers across all 4 models when the model has a non-degenerate baseline. The three exceptions (context_faithfulness, persona, fact_override) replicate as exceptions on all 4 models with the same sign. Pairwise Pearson correlations of $|\Delta_{\text{rand}}|$ across the 12-task set are: Llama-3B vs Qwen-3B $r = +0.77$, vs Llama-1B $r = +0.77$, vs Qwen-7B $r = +0.40$; Qwen-3B vs Qwen-7B $r = +0.64$. The Qwen-7B correlations are pulled down by one outlier: topic_suppression, where Qwen-7B has $\Delta_{\text{rand}} = -0.84$ while the other 3 models register near zero (see capability-ceiling discussion below). Excluding that single task, r with Qwen-7B rises to comparable values.

Capability ceiling cuts both ways. Several tasks confirm a *capability ceiling*: the geometric signature is identical across models, but the empirical effect depends on whether the model has the underlying capability or disposition for the steering to act on. **Downward:** Llama-1B's rule precision is 2/4 (refusal and sentiment steer as predicted; rag_distractor and hallucination_grounding are predicted to steer by geometry but show null effects). The two false positives are tasks where Llama-1B has near-zero baseline (rag_distractor baseline = 0.05, hallucination_grounding = 0.01), so the smaller model lacks the capability the steering could push on. **Upward:** Qwen-7B's topic_suppression shows $\Delta = -0.84$ despite var ≈ 0 (a rule miss) because Qwen-7B has a *strong* default disposition to refuse-to-discuss on our placeholder topics (baseline 0.84), and the structurally asymmetric contrast (refuse vs. discuss continuation) targets exactly that disposition. In both directions, the geometric signature recovers the same direction across models; what changes is whether the model encodes the corresponding behavior strongly enough for ablation to move it. A practitioner should pair the geometric rule with a quick baseline check on the held-out set.

B Layer stability of the rule (R2)

We profile per-layer $(\bar{\cos}, \sigma)$ for every (task, layer) pair on Llama-3B—all 28 layers, all 12 tasks, $N = 200$ pairs per task—using only forward passes (no generation cost). For each task we measure the layer-to-layer range of cos and var; the relevant quantity is whether the rule's classification of the task stays constant across layers.

Eight of twelve tasks are rule-stable at every layer. Honesty, sycophancy, sentiment, refusal, topic_suppression, context_faithfulness, persona, and politeness all classify the same way at every single one of the 28 layers (honesty / sycophancy / topic_suppression / context_faithfulness / persona / politeness are always low-var-collapsed; refusal is always live; sentiment is always live).

Four tasks show layer-dependent classification at extreme layers but agree at the mid-network choice ℓ=7. fact_override has $\bar{\cos} \in [0.02, 0.69]$ – the mean direction is unstable at every layer (rule classifies as "unstable mean" throughout, matching the main result). truthfulness, rag_distractor, and hallucination_grounding show some cosine variation across layers, but the rule classification is preserved at the mid-network layer choice we use in the main results. The cosine range for rag_distractor is $[0.69, 1.00]$, straddling the $\tau_{\text{stab}} = 0.7$ threshold; for truthfulness $[0.55, 0.80]$; for hallucination_grounding $[0.97, 1.00]$.

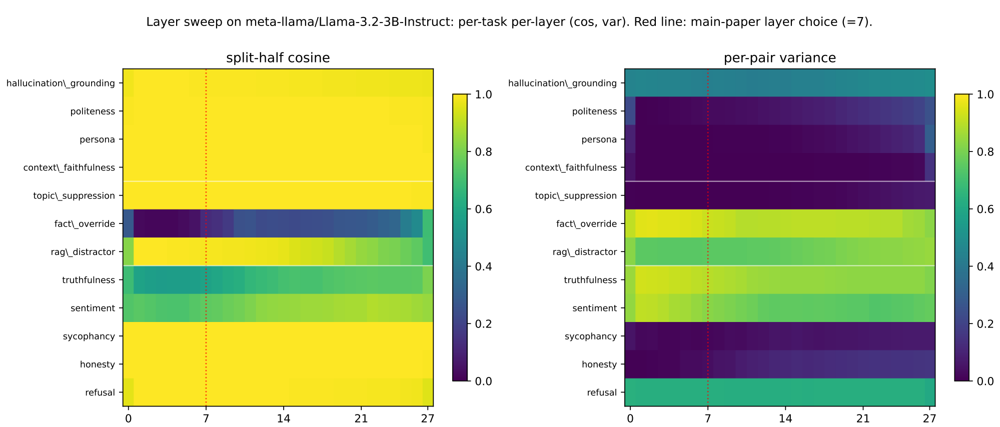

Figure 3: Layer sweep on Llama-3.2-3B-Instruct: per-task per-layer $(\bar{\cos}, \sigma)$. Rows are tasks (grouped by hypothesized kind, separated by white lines); columns are layers 0–27. Red dashed line: main-paper layer choice ($\ell = 7$). Row patterns do not change qualitatively between $\ell = 7$ and other mid-network layers, supporting the layer-stability claim of R2.

C Additive CAA (R3)

The main protocol uses scale-free ablation. To test that the predictive structure generalizes to additive CAA (Rimsky et al., 2023), we re-run the 4 representative tasks (refusal, rag_distractor, honesty, context_faithfulness) on Llama-3B at layer 7 under $h \leftarrow h + \alpha\bar{d}$ at every position, for $\alpha \in \{0.5, 1.0, 2.0, 4.0\}$. We use $N = 200$ pairs and $M = 50$ held-out examples per task (smaller than the main protocol to keep wall-clock manageable across 6 conditions per task).

	base	ablate	Δabl	α=.5	α=1	α=2	α=4
refusal	0.58	0.00	−0.58	0.62	0.64	0.60	0.52
honesty	0.44	0.42	−0.02	0.44	0.50	0.54	**0.62**
rag_distractor	0.56	0.00	−0.56	0.58	0.56	0.48	0.50
context_faithfulness	0.88	0.51	−0.37	0.90	0.90	0.81	0.74

Table 4: Ablation vs. additive CAA at layer 7 on Llama-3B, four representative tasks. Baseline (no intervention), scale-free ablation, and additive $h \leftarrow h + \alpha\bar{d}$ at four α values. Bold: cases where additive CAA produces an effect that ablation does not.

Sign and magnitude of ablation transfer to additive CAA on rule-predicted tasks. Both refusal and rag_distractor steer hard under ablation (-0.58 and -0.56). Under additive CAA they show non-monotonic patterns: a small enhancement at low α, then degradation back toward baseline (and beyond it) at high α. context_faithfulness behaves similarly: ablation -0.37; additive at $\alpha = 4$ gives -0.14. In all three rule-predicted cases, both interventions push the metric in the same direction; the magnitudes differ.

Additive CAA can succeed where ablation fails (honesty). Honesty's ablation effect is essentially null (-0.02), matching the main result. But *additive CAA at* $\alpha = 4$ *pushes the metric from* 0.44 *to* 0.62 (a $+0.18$ direction-specific increase in truthful answers, since the contrast is honest-vs-dishonest framing). The mean direction is real and behaviorally meaningful, but the model's natural behavior already lies orthogonal to it (removing it changes nothing), so only adding back along the direction produces an effect.

Implication for the rule. Our scale-free-ablation rule has high precision (every task it predicts to steer does steer under ablation) but may have a recall gap on low-variance contrasts that respond to additive but not ablative intervention. Honesty's behavior here suggests the rule could be sharpened by adding a second predictive condition for additive-CAA effects. We leave this to future work, noting that the rule's primary claim—predicting when scale-free ablation will move the metric—is unaffected.